Lab 3: Architectures and their costs

Reading and Learning Outcomes

Reading: Reddi, Chapter 4 DNN Architectures: §4.2 Multi-Layer Perceptrons, §4.3 Convolutional Neural Networks, §4.5 Attention Mechanisms, §4.6 Architectural Building Blocks, and §4.7 System-Level Building Blocks.

After completing this lab, students should be able to:

- Implement comparable MLP, CNN, and attention architectures in PyTorch.
- Calculate parameters and representative FLOPs by hand and compare with profiling tools.
- Measure inference latency for different batch sizes using a repeatable protocol.
- Explain architecture trade-offs using accuracy, parameter count, latency, and access patterns.

1. [10] Dataset: Fashion-MNIST in PyTorch. Use `torchvision.datasets.FashionMNIST`. Transform: `ToTensor()` and `Normalize((0.2860,), (0.3530,))`. Split official train 60k into 55k train and 5k validation. Use official 10k test. Batch size 128.
2. [25] Define three models of similar size, target 100k–150k parameters:
   - MLP: Flatten 784 → Linear 256 → ReLU → Linear 128 → ReLU → Linear 10.
   - CNN: Conv2d(1,16,3,padding=1) → ReLU → MaxPool2 → Conv2d(16,32,3,padding=1) → ReLU → MaxPool2 → Flatten → Linear(32×7×7,10).
   - Attention: split 28×28 image into 16 patches of 7×7. Linear projection to 64. Two Transformer encoder layers: `d_model=64`, `nhead=4`, `dim_feedforward=128`. Mean pool. Linear 64→10.
3. [25] Compute cost:
   - Count parameters with `torchinfo.summary`.
   - Hand-calculate FLOPs for one layer of each type using Ch. 4 formulas. Linear: `2 × in × out`. Conv2d: `2 × K × K × C_in × C_out × H_out × W_out`. Attention: QKV projections `2 × seq × d_model × 3 × d_model`, attention scores `2 × seq² × d_model`, etc.
   - Compare with `thop.profile`. Explain any gap.
   - Measure inference latency for batch 1 and batch 128. Median of 100 and 20 runs.
4. [30] Train all three for 5 epochs with Adam, lr `1e−3`, batch 128. Record test accuracy.
5. [15] Plot accuracy against parameters and against latency. Use Ch. 4.6–4.7 access patterns to explain which architecture is most cost-effective.
6. [10] Write `report.md` and commit.

Graded evidence: hand-calculated vs measured FLOPs, trade-off plots, written explanation.